Poisson Distribution

What is a Poisson Process?

- The average time between events is known ($E[X] = \lambda$) and constant
- Events are independent of each other (memoryless property)
- Events cannot occur simultaneously, and the time between two events is a Bernoulli trial.

Comparison to the Exponential Distribution

	Exponential	**Poisson**
Question	How much *time* between a given number of events?	How many events occur in a time interval?
Random Variable	Time (continuous)	Number of events (discrete)
Parameters	λ (rate of occurrence), $[\lambda] = s^{-1}$	λ (expected number of occurrences), unitless
Notes	The expontential distribution is a special case of the gamma distribution where the shape parameter is 1 and the scale parameter is $1/\lambda$. It is also the continuous analog for the geometric distribution	-

References

- Understanding Exponential vs Poisson Distributions